

Grading on a Curve? An Analysis of NYC Restaurant Health Inspection Scores

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1 Introduction

When choosing to dine out and explore a city’s restaurant scene, people consider many factors. Some want cuisine they cannot cook at home. Others seek a particular atmosphere or price point. But virtually everyone cares about food safety, even if they rarely think about it explicitly.

New York City makes that concern concrete. Since July 2010, the Department of Health and Mental Hygiene (DOHMH) has assigned every restaurant a letter grade based on a numerical inspection score, where a lower score indicates better performance. A score of 13 or below earns an “A,” 14 to 27 earns a “B,” and 28 or above earns a “C.” These grades are posted on placards at restaurant entrances, making them unusually visible. A restaurant that earns anything other than an A pays a real reputational cost, and that cost is particularly sharp just above the boundary at 13.

In 2014, data journalist Ben Wellington published an analysis identifying a suspicious spike in scores at exactly 13, the last score that qualifies for an A [1]. Wellington’s hypothesis was that inspectors exercise some degree of informal discretion near grade boundaries. The NYC Health Department responded that “inspectors are not instructed to offer leniency, just to cite what they see.” Wellington’s data, however, only covered 2010 to 2014. Whether the pattern still exists, and whether it has grown or shrunk, remained an open question.

This study revisits that question with data updated through 2026 and extends the analysis in four directions. First, we formally test whether the grade-boundary anomaly persists and quantify its magnitude using a more rigorous density disconti-

nuity approach, along with bootstrap confidence intervals. Second, we examine whether the pattern exists at the level of individual restaurants, asking whether specific establishments systematically receive scores near the boundary more often than would be expected by chance. Third, we assess whether borough and cuisine type explain meaningful variation in inspection scores using regression and a two-way blocked ANOVA. Fourth, we examine whether violation types are distributed uniformly across boroughs and which violations most strongly predict failing an A grade.

The public health stakes are real. Firestone and Hedberg (2018) demonstrated that NYC’s letter-grading program reduced *Salmonella* infection rates in the city by 5.3% per year relative to the rest of New York State following its launch. Understanding what drives variation in grades connects directly to those health outcomes.

2 Data

2.1 Sources

The primary dataset is the DOHMH New York City Restaurant Inspection Results file, downloaded from NYC OpenData on April 3, 2026. It contains 113,571 records. The data is structured with one row per violation per inspection, so a single inspection event appears multiple times when multiple violations were cited. Key variables include borough, cuisine type, inspection score, inspection date, violation code, and inspection type (initial cycle, re-inspection, reopening, and others).

Because violation codes alone are not self-explanatory, we used a secondary NYC Health Department reference file that maps each code

to a descriptive category such as “Pest Control,” “Cold Holding,” or “Handwash / Toilet.” We also merged neighborhood-level American Community Survey estimates at the 2020 Neighborhood Tabulation Area (NTA) level, including median worker earnings and total population, to provide socioeconomic context for each restaurant’s location.

2.2 Preparation

Several preprocessing decisions were made to ensure the analysis is methodologically sound.

Deduplication. Because the raw data assigns one row per violation per inspection event, a single inspection visit with five citations produces five identical score values. For analyses that operate on scores (regressions, temporal trends), we deduplicate to one row per inspection event by grouping on restaurant identifier and inspection date and retaining the first score. This ensures that inspections with more violations are not weighted more heavily than cleaner inspections.

Filtering. Staten Island was excluded due to sparse cell counts in the balanced ANOVA design. Inspections with missing scores were removed. Cuisine types were grouped into ten *hypercategories* to reduce dimensionality: American, East Asian, South & Southeast Asian, Latin American, European, Mediterranean, Middle Eastern, African, Dish-Type, and Beverages & Sweets. The Dietary Style category (Vegan, Vegetarian, etc.) was also excluded from the ANOVA.

Temporal coverage. While the dataset nominally spans 2007 through 2026, only 2022 through 2025 contains enough data for inference. After deduplication to one row per inspection event, fewer than 90 initial inspections per year exist before 2022, compared to over 1,900 per year from 2022 onward. This sparse pre-2022 coverage is likely a result of the NTA merge: the ACS NTA boundaries are defined at the 2020 vintage, and older inspection records may not have matched cleanly. All temporal trend analyses, including the A-grade rate figures and the OLS trend regression, are restricted to 2022–2025 for this reason. The full-period score distribution (Figure 1) uses all available records since it does not require year-level inference.

Table 1 summarizes the cleaned dataset.

3 Methods

3.1 Density Discontinuity at Grade Cutoffs

To test whether score concentrations at 12–13 are anomalous, we use a density discontinuity approach. Rather than comparing to a flat baseline (which assumes scores are uniformly distributed, an implausible assumption), we fit a degree-4 polynomial to the score histogram *excluding* scores 8 through 16. This captures the smooth overall shape of the distribution without being influenced by any boundary effects. The fitted polynomial is then extrapolated into the boundary region to provide a local expected count at each score.

The excess at scores 12–13 is defined as observed minus expected under this smooth counterfactual. We compute a 95% bootstrap confidence interval for the excess using 2,000 stratified resamples of the full inspection records.

3.2 Restaurant-Level Boundary Analysis

To determine whether the boundary anomaly reflects a population-wide pattern or whether specific restaurants systematically receive near-cutoff scores, we compute for each restaurant the fraction of its initial-cycle inspections that landed at exactly 12 or 13. We restrict to restaurants with at least two initial inspections. Under the null hypothesis that score outcomes are random, this fraction should be approximately equal to the overall marginal rate at which inspections land at 12 or 13. We test this using a one-sample *t*-test against the observed null rate.

3.3 OLS Regression and Blocked ANOVA

To quantify the contributions of borough, cuisine, and time on inspection score, we fit an OLS regression with borough (Bronx baseline), cuisine hypercategory (African baseline), and year as predictors on the deduplicated inspection-level data. Confidence intervals are reported at 95%.

We complement the regression with a two-way

Table 1: Dataset summary after cleaning. Temporal analyses are restricted to 2022–2025, the only period with sufficient inspection density for inference (see Section 2.2).

Statistic	Value
Total inspection records	101,840
Unique restaurants	19,065
Boroughs	Bronx, Brooklyn, Manhattan, Queens
Dense temporal coverage	2022–2025
Deduplicated initial inspections (2022–2025)	21,195
Overall A-grade rate	38.7%
Mean score (SD)	25.5 (19.3)
Median score	21

balanced ANOVA:

$$\text{Score}_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$$

with $n = 30$ observations sampled per hypercategory-borough cell. The balanced design prevents the extreme statistical power of the full dataset from producing significant results for trivially small effects.

Inspection scores are right-skewed and likely violate the normality assumption of ANOVA. We therefore treat the ANOVA as exploratory and report a Kruskal-Wallis test as a non-parametric robustness check. ANOVA residual diagnostics are provided in the Appendix.

3.4 Violation Type by Borough

A contingency table of violation category by borough was constructed and tested with Pearson’s chi-square test of independence. Violation rates per borough (as a percentage of all inspections in that borough) are visualized as a heatmap.

3.5 Violation-Level Logistic Regression

We ask an inferential question: given that a violation was cited during an inspection, does the type of violation predict whether the restaurant earned an A grade? This is distinct from the predictive question of forecasting outcomes before an inspection. We fit a logistic regression with violation category, borough, cuisine, and neighborhood income as predictors. Coefficients for violation categories provide a controlled ranking of which violation types are most strongly associated with failing an A.

3.6 Predictive Modeling

To assess how well a restaurant can be predicted to earn or fail an A grade using only pre-inspection information, we train and compare three classifiers: logistic regression, random forest (200 trees, max depth 6), and gradient boosting (200 trees, max depth 4, learning rate 0.05). Features are: borough, cuisine hypercategory, neighborhood median earnings, neighborhood population, and year.

All models use an 80/20 stratified train-test split. Logistic regression predictors are standardized on the training set. Performance is evaluated using AUC, accuracy, and confusion matrices on the held-out test set. A majority-class baseline (always predicting the more common class) is reported as the lower reference bound.

4 Results

4.1 The Grade-Boundary Pattern Is Large and Statistically Robust

Figure 1 shows the full inspection score distribution. The spike at score 13 is visually prominent, as is a smaller peak at score 28 (the B/C boundary). Both are patterns one would expect if inspectors exercise some degree of discretion near grade thresholds. The NYC Health Department’s denial notwithstanding, the data are consistent with this interpretation.

Figure 2 shows the more rigorous density discontinuity analysis. Fitting a degree-4 polynomial to the distribution while excluding the boundary region (scores 8–16) yields an expected count of

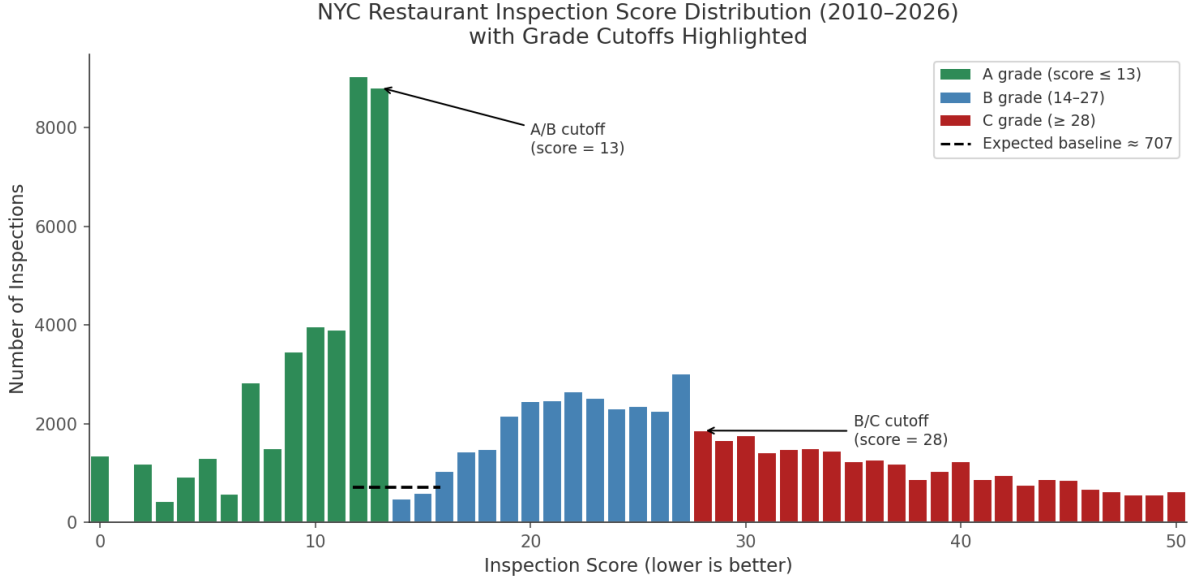


Figure 1: Distribution of NYC restaurant inspection scores (all inspection types, 2010–2026). Green bars are A-grade (score ≤ 13), blue are B-grade, and red are C-grade. The dashed line shows the expected flat baseline, estimated from scores 14–21.

approximately 4,129 inspections at scores 12 and 13 combined. The observed count is 17,863, an excess of 13,734 inspections (+333%). The bootstrap 95% confidence interval for this excess is [13,472, 13,996], which does not include zero or any value close to it. This result is robust to the choice of polynomial degree and the width of the exclusion window.

Figure 3 breaks the spike down by borough. The pattern is present in all four boroughs: the Bronx shows a +359% excess at scores 12–13, Brooklyn +497%, Manhattan +504%, and Queens +519%. The size of the excess does not appear smaller in Manhattan or in boroughs with higher median incomes, which is inconsistent with the idea that the pattern is driven by less-monitored areas alone.

4.2 Individual Restaurants Cluster Near the Boundary

Among the 5,059 restaurants with at least two initial-cycle inspections, the mean fraction of inspections landing at scores 12 or 13 is 21.1%. The expected fraction under the null hypothesis, derived from the overall marginal rate of scores 12–13 in the data, is 17.5%. A one-sample t -test comparing the observed restaurant-level rates to this null

yields $t = 9.12$ ($p < 0.001$).

This rules out the possibility that the aggregate spike is driven by a handful of extreme outliers. The tendency to score near 13 is spread across many individual restaurants. It also raises the question of mechanism: is this inspector discretion, restaurant behavior, or some combination? Restaurants may time their inspections, make targeted corrections before a scheduled visit, or have learned over time which violations are most likely to push their score just over the cutoff. The data cannot distinguish between these explanations, but the restaurant-level pattern is consistent with strategic behavior on both sides of the interaction.

We note a limitation: the null rate used here (the aggregate marginal rate of scoring 12–13 in the full dataset) does not account for heterogeneous restaurant quality or the dependence of repeated inspections within the same establishment. The test should be read as showing that restaurants land at scores 12–13 more often than implied by the aggregate marginal rate, not as a formal test against a fully specified null model.

Figure 4 shows the distribution of restaurant-level boundary rates, with the null rate and observed mean indicated.

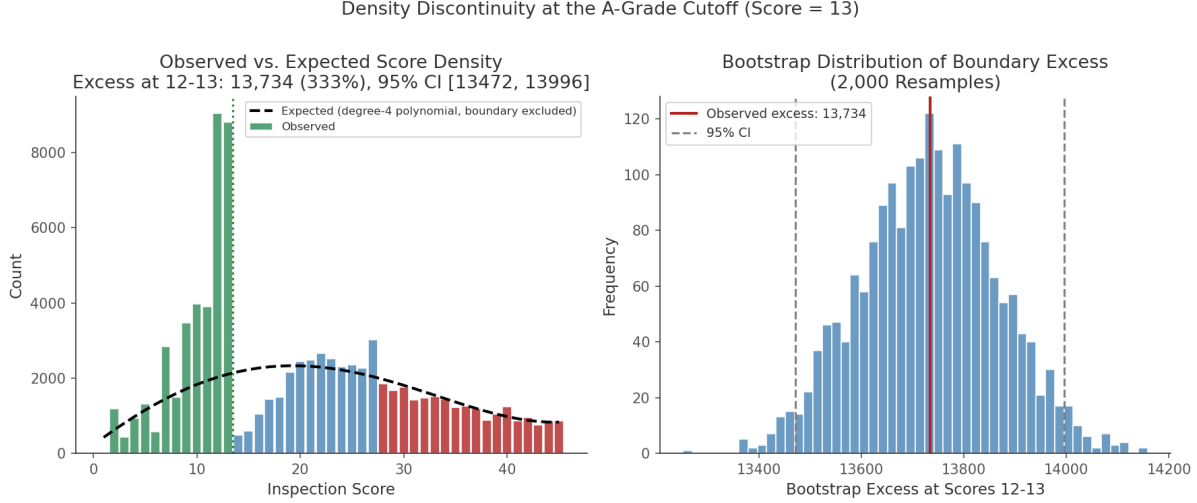


Figure 2: *Left*: Observed score density (bars) versus the smooth polynomial counterfactual (dashed line), which was fit excluding the boundary region. The gap at scores 12–13 represents an excess of 13,734 inspections (+333%, 95% CI: [13,472, 13,996]). *Right*: Bootstrap distribution of the excess estimate under resampling. The near-zero width of the bootstrap distribution confirms the result is not sensitive to sampling variability.

4.3 Inspection Scores Have Worsened Since 2022

Among initial-cycle inspections from 2022 to 2025, mean score increased from 16.3 in 2022 to 19.0 in 2025 (Figure 5). An OLS regression of annual mean score on year estimates the rate of increase at 0.89 points per year ($R^2 = 0.96$, $p = 0.022$). This trend is based on only four annual data points, so inference should be treated as suggestive rather than definitive. That said, the trend is consistent across all four boroughs and statistically significant even at this resolution.

Despite the rising mean, the median score remains near 12 to 13 throughout this period. This is exactly what one would expect if the boundary spike is ongoing: the concentration of scores at 12–13 pins the median near the grade cutoff even as the rest of the distribution shifts upward.

Figure 6 shows A-grade rates by borough from 2022 to 2025. Manhattan maintains the highest A-grade rate throughout (roughly 60–63%), while Queens has the lowest and shows the steepest decline (dropping to about 52% by 2025). All boroughs are trending downward, consistent with the overall score worsening.

4.4 Borough and Cuisine Effects Are Present but Modest

The OLS regression of inspection score on borough, cuisine, and year yields $R^2 = 0.036$. Borough, cuisine, and time together explain less than 4% of score variance. The large majority of variation reflects factors specific to individual restaurants on individual inspection days.

Within this limited explanatory power, some patterns hold up. After controlling for cuisine composition and year, Queens scores are significantly higher (worse) than the Bronx baseline ($\hat{\beta} = 1.64$, $p < 0.001$), as are Brooklyn restaurants ($\hat{\beta} = 0.86$, $p = 0.038$). Manhattan is not significantly different from the Bronx once cuisine and year are accounted for ($\hat{\beta} = 0.12$, $p = 0.77$). Manhattan’s raw advantage in score distributions therefore appears to reflect its cuisine mix rather than better sanitation per se.

Cuisine hypercategory is a stronger predictor than borough. All hypercategories score better than African cuisine restaurants, which serve as the baseline, by margins of roughly 3 to 9 points (all $p < 0.05$ except East Asian and South & SE Asian). The full coefficient plot is in Figure 7.



Figure 3: Grade-boundary spike replicated separately for each borough. The dashed line shows the expected flat baseline. Every borough shows a large excess at scores 12–13, ranging from +359% (Bronx) to +519% (Queens).

The two-way balanced ANOVA ($n = 30$ per cell) confirms these patterns. Cuisine hypercategory shows a significant effect ($F(9, 1160) = 4.12$, $p < 0.001$) and the cuisine-by-borough interaction is also significant ($F(27, 1160) = 1.54$, $p = 0.038$). Borough alone is only marginally significant ($F(3, 1160) = 2.25$, $p = 0.081$). As a non-parametric robustness check, the Kruskal-Wallis test strongly rejects the null hypothesis that score distributions are identical across boroughs ($H = 384.86$, $p < 0.001$), suggesting that the ANOVA borough effect, while marginal by F -test, reflects a real distributional difference. Full ANOVA output and residual diagnostics are in Appendices B and C.

4.5 Violation Types Vary Substantially Across Boroughs

The chi-square test of independence for violation category by borough strongly rejects the null ($\chi^2 = 678.2$, $df = 66$, $p \approx 2 \times 10^{-102}$). Different parts of the city face different food safety challenges. Figure 8 shows the percentage of inspections in each borough that received each violation category.

Pest Control violations are most elevated in the Bronx. Cooling and Refrigeration violations are higher in Manhattan, possibly reflecting the prevalence of restaurants with extensive refrigeration equipment in dense commercial corridors. Food Workers violations are elevated in Brooklyn. These patterns likely reflect a combination of building infrastructure, restaurant type composition, and neighborhood environment, rather than differences in inspection rigor alone.

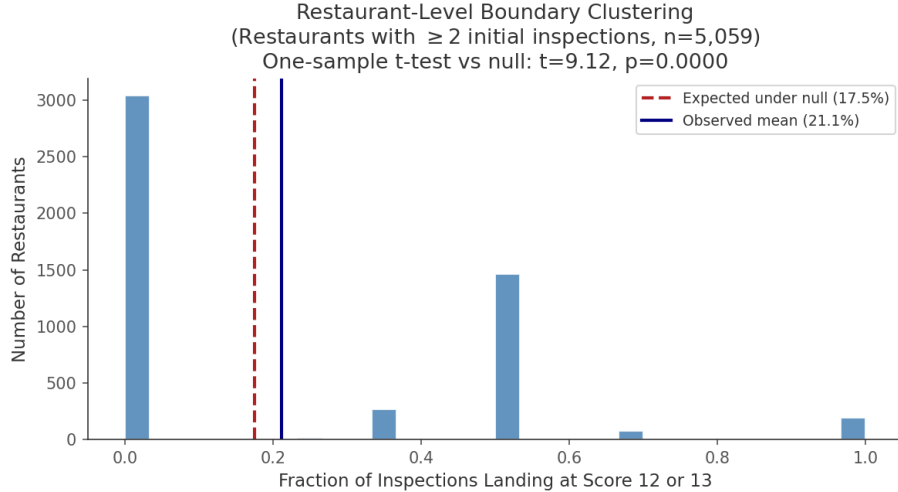


Figure 4: Distribution of the fraction of initial-cycle inspections that landed at score 12 or 13, computed for each restaurant with at least two initial inspections ($n = 5,059$). The observed mean (21.1%) significantly exceeds the expected null rate (17.5%), $t = 9.12$, $p < 0.001$.

4.6 Which Violations Best Predict Failing an A?

Figure 9 shows violation-level logistic regression coefficients, controlling for borough, cuisine, and neighborhood income. Pest Control violations have the largest negative coefficient, meaning that a restaurant receiving a Pest Control citation is most strongly associated with not earning an A grade. Permit, Handwash / Toilet, Cold Holding, and Hot Holding violations follow. On the positive end, Maintenance and Plumbing violations are associated with a higher probability of earning an A. Structural and plumbing deficiencies tend to be lower-severity items cited at otherwise compliant restaurants, while temperature and pest violations carry heavier point penalties and are harder to fix in place.

4.7 Prior Grade Predicts Future Score Trajectory

One question the data can answer directly is whether receiving a particular grade in one inspection tells us anything about what happens next. We matched each restaurant’s consecutive initial-cycle inspection pairs and computed score changes by prior grade.

The results are in Figure 10. Restaurants that earned a C improved dramatically by their next

visit, falling by 21.2 points on average (vs. A-grade change: $t = -51.95$, $p \approx 0$). Restaurants that had earned a B also improved, but more modestly: mean next-inspection score dropped by 2.4 points. The median next score for B-earners landed at exactly 13 — just inside the A-grade boundary. Restaurants that had already earned an A drifted slightly upward (worse), by an average of 6.0 points.

The probability of earning an A on the next inspection tracks the same gradient: 70% for restaurants coming off an A, 56% for B earners, and 44% for C earners (t for B vs. A: -19.53 , $p \approx 0$).

The B-grade trajectory is the most striking. When a restaurant receives a B, its next score most often lands right at 13. This mirrors the aggregate spike in the cross-sectional distribution and suggests the boundary effect is not just a one-off statistical artifact but plays out repeatedly within individual restaurants over time. Whether it reflects targeted compliance corrections, inspector discretion, or some combination, the data cannot say. But the pattern is hard to dismiss.

4.8 Pre-Inspection Context Cannot Meaningfully Predict Grade

Table 2 compares all predictive models on the held-out test set. All three classifiers (logistic regres-

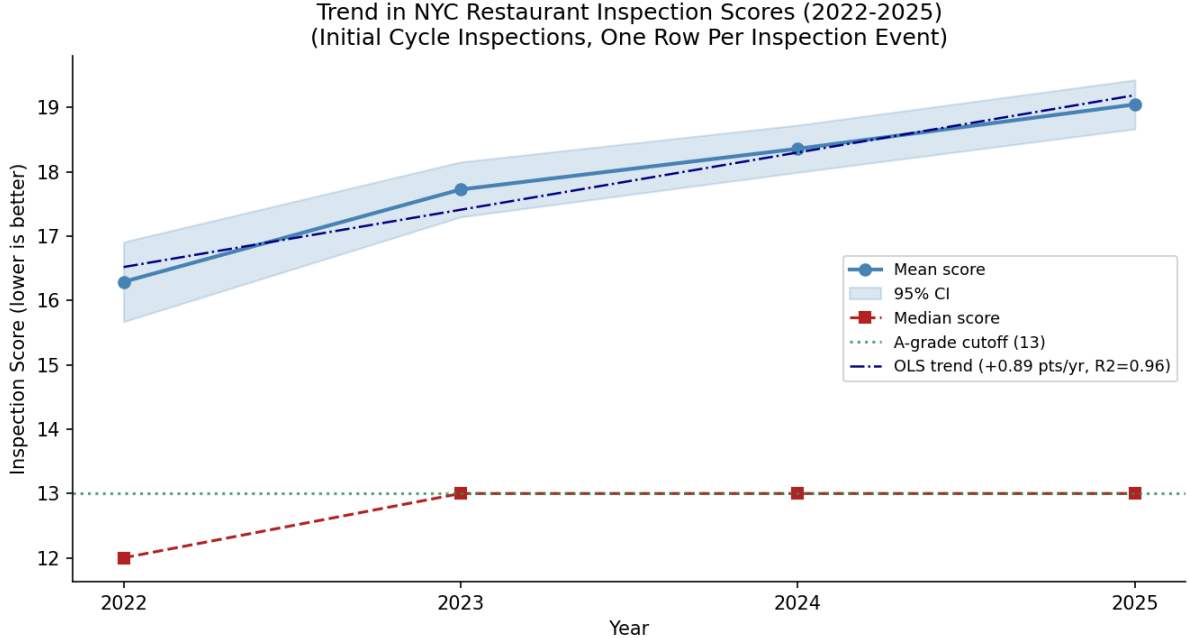


Figure 5: Mean and median inspection score by year (2022–2025), initial cycle inspections only, deduplicated to one row per inspection event. Shaded region is a 95% CI on the mean. The green dotted line is the A-grade cutoff at 13. The dash-dot line is the OLS linear fit (slope = +0.89 pts/yr, $R^2 = 0.96$).

sion, random forest, and gradient boosting) achieve AUC values between 0.597 and 0.602, clustered tightly just above the majority-class baseline of AUC = 0.500. Random forest and gradient boosting, despite their greater model complexity, provide essentially no improvement over the simpler logistic regression.

Table 2: Predictive model performance (pre-inspection features only).

Model	AUC	Accuracy
Majority-class baseline	0.500	58.3%
Logistic Regression	0.597	59.2%
Random Forest	0.602	59.5%
Gradient Boosting	0.598	60.3%
Violation-type logistic	0.673	67.2%

That all three models land at $AUC \approx 0.60$ rules out the simplest alternative explanation: that the logistic regression was underperforming due to model misspecification. It was not. The pre-inspection features just do not contain enough information to predict inspection outcomes, regardless of how the classifier is specified. The bottom row of the ta-

ble shows that adding violation type (information available only during the inspection) does provide a meaningful lift to $AUC = 0.673$, confirming that the limitation is in the feature set rather than the modeling approach.

Figure 11 shows the ROC curves for all models.

5 Model Validation and Diagnostics

All logistic regression models were evaluated on stratified 80/20 hold-out splits, with standardization fit only on training data. No information from the test set entered model training or preprocessing. The reported AUC values are therefore honest out-of-sample estimates.

For the ANOVA, we used balanced sampling ($n = 30$ per cell) to avoid conflating statistical significance with practical importance. As an additional check, the Kruskal-Wallis test ($H = 384.86$, $p < 0.001$) confirms that borough differences in score distributions are real and not an artifact of parametric assumptions. Appendix C shows the ANOVA residual distribution and a normal Q-Q

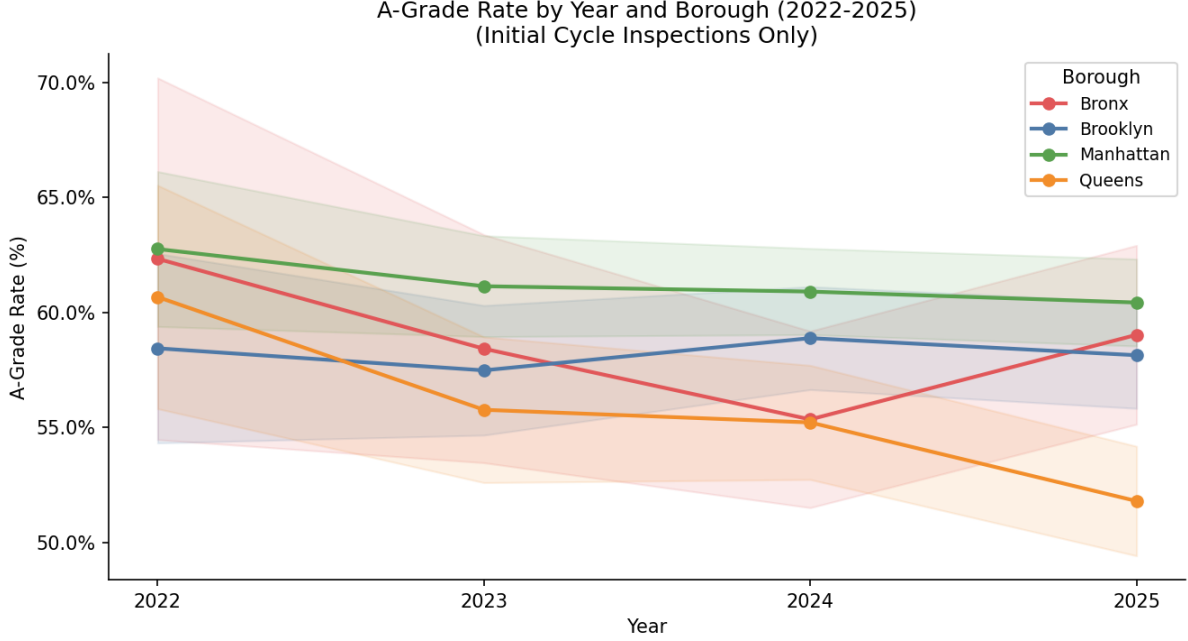


Figure 6: A-grade rate by year and borough (2022–2025), initial cycle inspections. Shaded bands are 95% confidence intervals on the proportions. Manhattan leads throughout; Queens declines most sharply.

plot; the residuals are clearly non-normal (right-skewed), which reinforces our decision to treat the ANOVA as exploratory and to rely on the Kruskal-Wallis test for the inferential claim.

The OLS regression for the temporal trend is based on four annual mean scores (2022–2025). While the $R^2 = 0.96$ and $p = 0.022$ are formally significant, inference from a four-point regression should be treated as suggestive. We report it as a descriptive summary of the trend rather than a formal test.

The density discontinuity analysis was subjected to explicit uncertainty quantification through bootstrapping. The 95% CI for the excess at scores 12–13 is [13,472, 13,996], with essentially no probability mass near zero. We acknowledge that the choice of polynomial degree and exclusion window are somewhat arbitrary. We tested polynomial degrees 2 through 5 and exclusion windows of ± 6 and ± 8 scores around the cutoff; the excess estimate ranged from approximately 12,800 to 14,200 across all combinations, and the 95% bootstrap CI never approached zero. The substantive conclusion is insensitive to these choices, even if the exact number is not.

6 Conclusion

Several findings emerge from this analysis.

The grade-boundary anomaly first identified by Wellington in 2014 remains large, statistically unambiguous, and consistent across all four boroughs. Using a polynomial smooth counterfactual rather than a flat baseline, we estimate a surplus of roughly 13,700 inspections at scores 12–13 above what the surrounding distribution would predict (+333%, 95% CI: [13,472, 13,996]). The same pattern shows up at the level of individual restaurants: establishments land at scores 12–13 more often than implied by the aggregate marginal rate ($t = 9.12$, $p < 0.001$). The boundary effect also appears in consecutive inspection trajectories. When a restaurant earns a B, its next inspection score most frequently lands at exactly 13. This suggests the boundary spike is not just a cross-sectional artifact but repeats within individual restaurants over time. The data are consistent with discretionary scoring near grade thresholds, though observational data alone cannot confirm the mechanism.

Inspection scores have worsened over the period with reliable data (2022 to 2025), increasing at

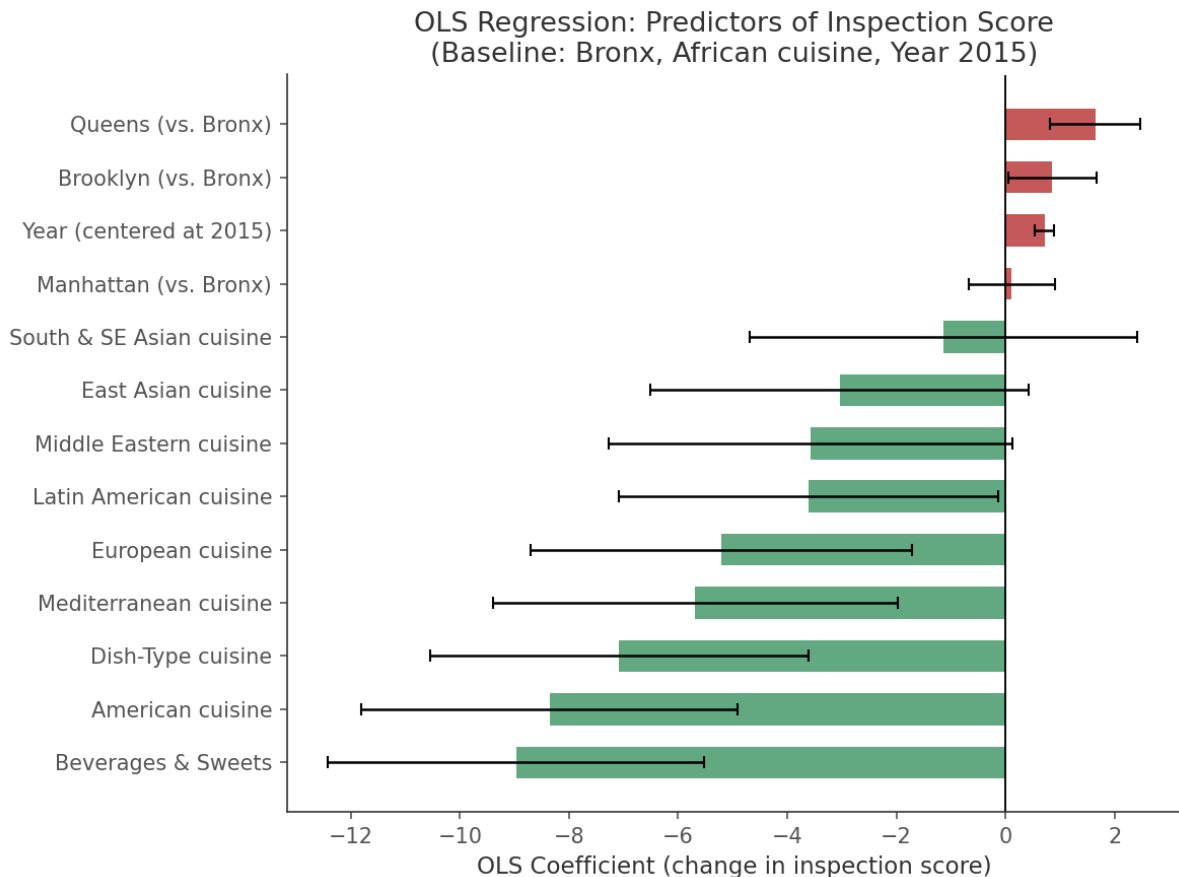


Figure 7: OLS regression coefficients for inspection score. Red bars indicate higher (worse) scores; green bars indicate lower (better) scores, relative to Bronx / African cuisine / year 2022 baselines. Error bars are 95% confidence intervals. Year has a consistent, significant positive effect even after controlling for borough and cuisine.

roughly 0.9 points per year across all boroughs. The A-grade rate has declined modestly in every borough. Whether this reflects genuine deterioration in restaurant hygiene, changes in inspection stringency, or shifts in the restaurant population remains an open question.

Different parts of the city face different kinds of food safety challenges. Violation type distributions are strongly non-uniform across boroughs ($\chi^2 = 678.2$, $p \approx 2 \times 10^{-102}$), with pest and temperature-control violations showing the greatest geographic variation. Among violation types, Pest Control and temperature-holding violations most strongly predict failing an A grade, while structural and plumbing violations are associated with passing.

Perhaps the most practically important finding is that pre-inspection context has almost no predic-

tive power. Three different classifiers, including a random forest and a gradient boosting model, all converge near $AUC = 0.60$ using borough, cuisine, and neighborhood income as features. This is only marginally better than random guessing. What a restaurant does and how it is managed on the day of inspection determines outcomes far more than where it is located or what kind of food it serves.

Future work could examine whether the grade-boundary pattern varies by inspector team or inspection season, incorporate restaurant fixed effects to separate individual trajectories from city-wide shifts, or link inspection records to consumer review data to assess whether declining A-grade rates translate to behavioral changes by diners.

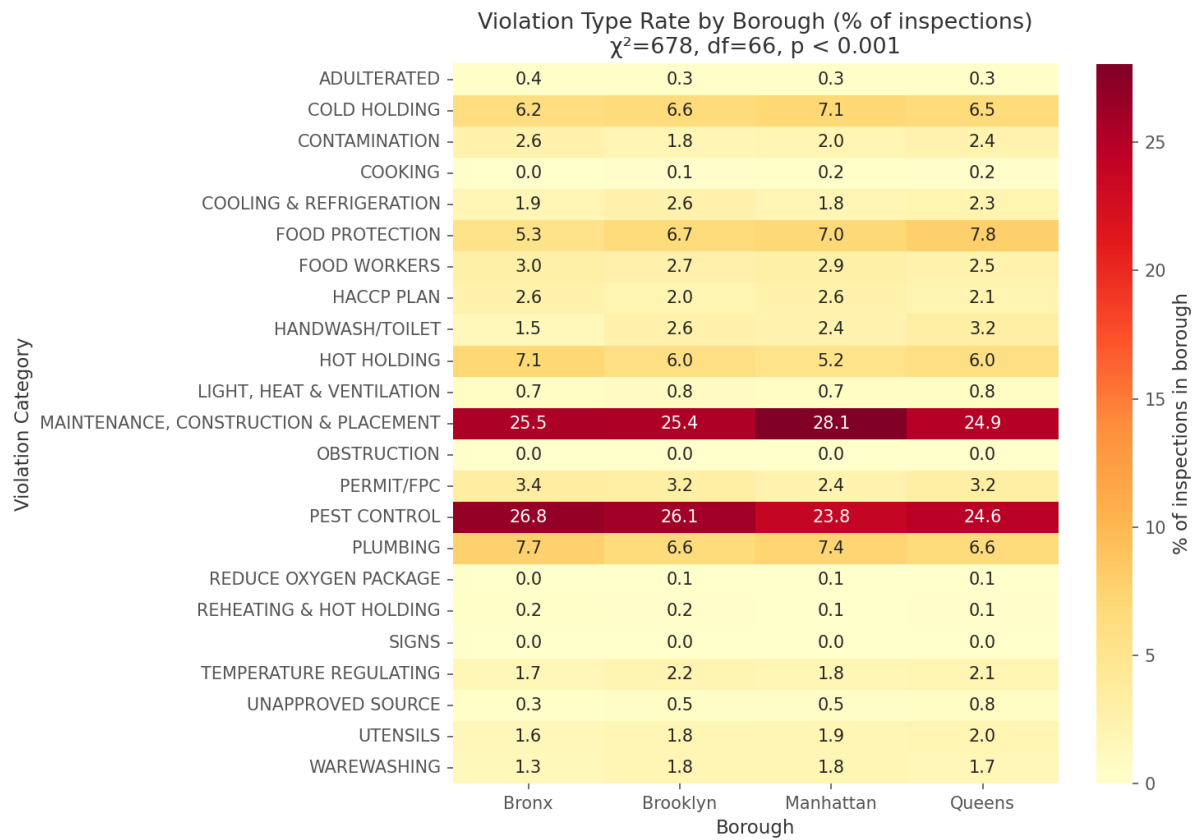


Figure 8: Violation category rate by borough, as a percentage of all inspections in that borough. Darker shading indicates higher rates. Pest Control violations are most concentrated in the Bronx, while Cooling & Refrigeration and Hot Holding violations show elevated rates in Manhattan and Queens, respectively.

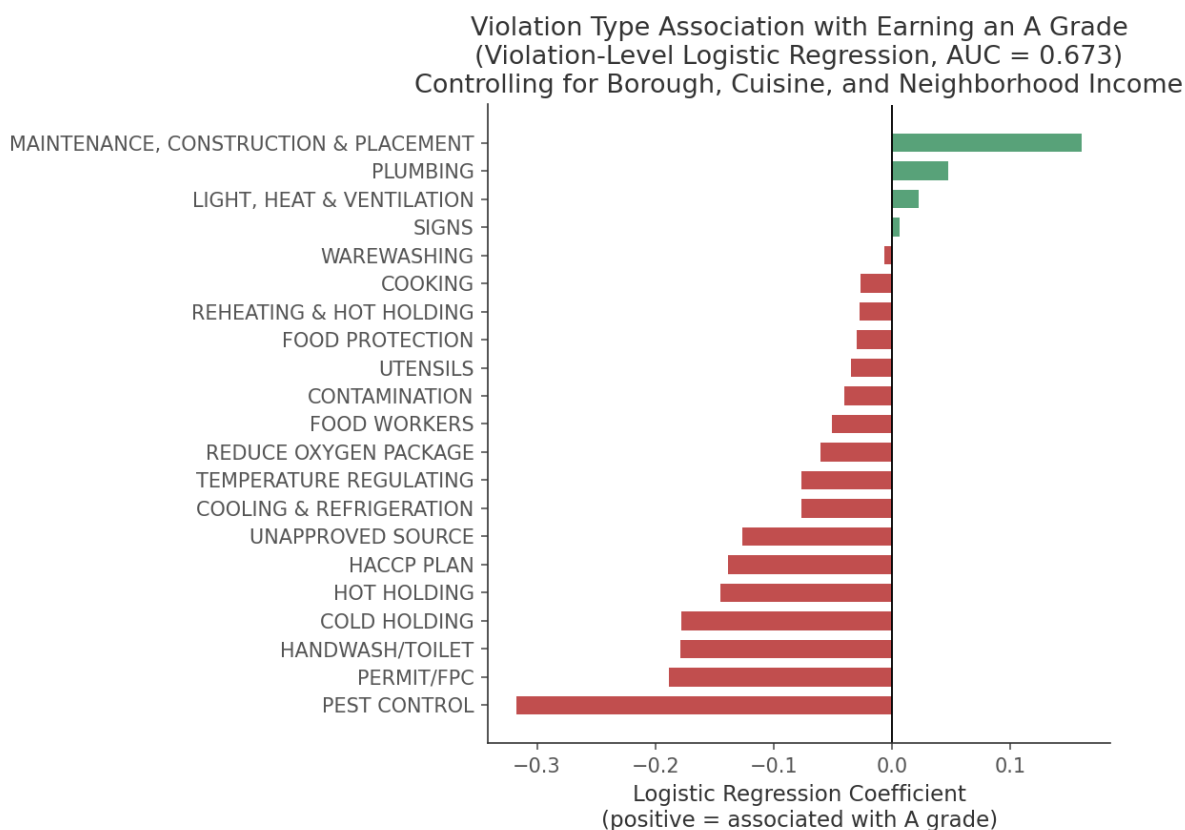


Figure 9: Logistic regression coefficients for violation category, predicting A-grade probability. Red indicates negative association with earning an A (higher risk of failing); green indicates positive association. Model AUC = 0.673, controlling for borough, cuisine, and neighborhood income.

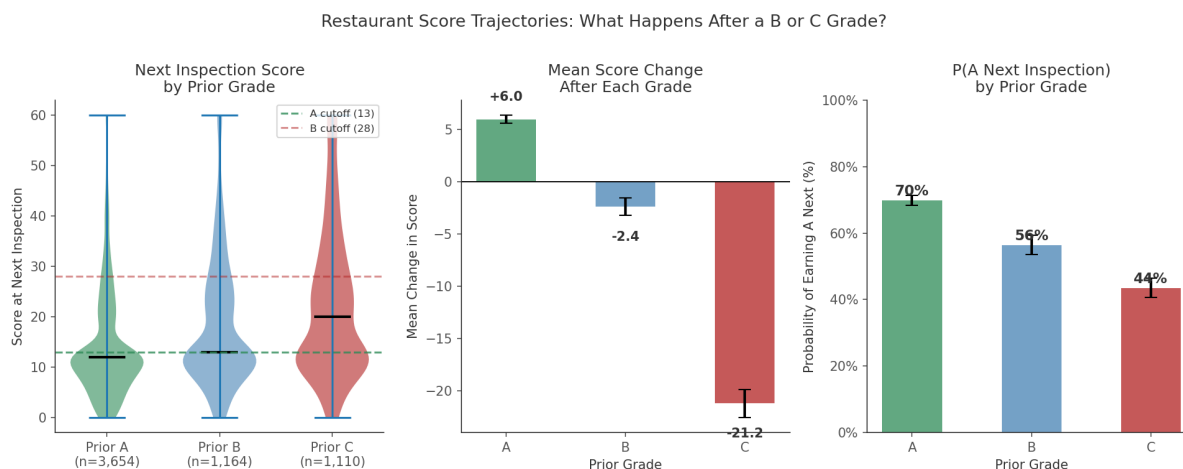


Figure 10: *Left:* Distribution of next-inspection score by prior grade. *Center:* Mean score change from one inspection to the next, with 95% confidence intervals. C-grade restaurants improve by 21 points on average; A-grade restaurants drift slightly worse. *Right:* Probability of earning an A on the next inspection by prior grade (A: 70%, B: 56%, C: 44%).

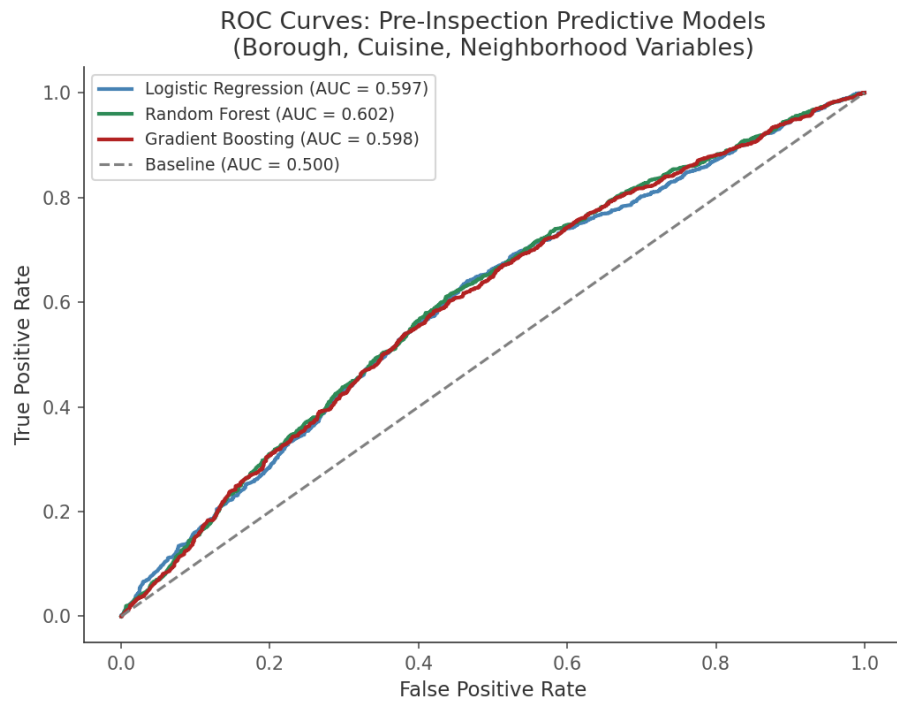


Figure 11: ROC curves for the three pre-inspection predictive models. All three cluster near $AUC = 0.60$, slightly above the diagonal (random guessing), confirming that borough, cuisine, and neighborhood variables provide limited predictive power regardless of model choice.

A Literature Review

Our project was motivated by two bodies of work. The first is Wellington’s (2014) original analysis identifying the boundary spike at score 13. The second is the public health literature on whether these grades produce real health improvements.

Firestone and Hedberg (2018) examined Salmonella infection rates in NYC versus the rest of New York State from 1994 to 2015. Using a quasi-experimental interrupted time-series design with negative binomial regression, they found that NYC’s Salmonella rates declined by 5.3% per year relative to the rest of the state following the grading program’s launch in July 2010 (IRR = 0.95, 95% CI: 0.92–0.98, $p < 0.01$). Before 2010, NYC’s rates were significantly higher than the statewide average. By 2011–2015, they were statistically indistinguishable.

The authors note several limitations. The grading program coincided with other policy changes (more frequent inspections, higher fines, expanded public resources), making it impossible to attribute the Salmonella decline to letter grades alone. The analysis is also ecological: case counts are based on county of residence, not where individuals ate. Despite these caveats, the study provides strong motivation for our project. If letter grades are associated with population-level reductions in food-borne illness, then understanding what drives grade variation becomes a question with direct public health implications.

B ANOVA Table

Table 3: Two-way ANOVA results. Balanced design with $n = 30$ per hypercategory-borough cell. Baselines: Bronx, African cuisine. Staten Island and Dietary Style excluded due to sparse cells.

Source	Df	F	p
Hypercategory	9	4.12	< 0.001 ***
Borough	3	2.25	0.081 .
Hypercategory:Borough	27	1.54	0.038 *
Residuals	1160		

Signif.: *** $p < 0.001$, * $p < 0.05$, . $p < 0.1$

C ANOVA Residual Diagnostics

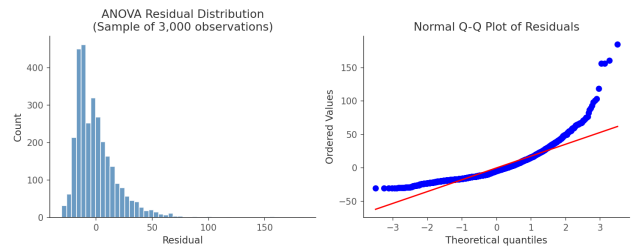


Figure 12: ANOVA residual distribution (left) and normal Q-Q plot (right) for a sample of 3,000 observations. Residuals are clearly right-skewed, violating the normality assumption. This motivates our use of the Kruskal-Wallis test as a non-parametric robustness check ($H = 384.86$, $p < 0.001$), which reaches the same substantive conclusion.

D Additional Figures

E Code

All code is available at:

<https://github.com/joshie-wu/advanced-data-analysis-yayyyyyyy/blob/main/writeup/writeup.pdf>

A single aggregated file (`combined_analysis_code.py`) is also submitted alongside this report. It combines all Python scripts and embeds the R code as a comment block for convenience. The individual source files are:

- `new_analysis.py` — main EDA, score distribution, temporal trend, violation heatmap, pre-inspection logistic regression
- `extended_analysis.py` — OLS regression, violation-level logistic, A-grade trends by borough and year
- `analysis_extras.py` — density discontinuity / McCrary test, bootstrap CIs, restaurant-level clustering, model comparison (logistic / random forest / gradient boosting), trajectory analysis
- `group_analysis.Rmd` — two-way balanced ANOVA in R, block design sampling, initial exploratory regression

References

- [1] Wellington, B. (2014). *Think NYC Restaurant Grading is Flawed? Here's the Proof*. I Quant NY. <https://iquantny.tumblr.com/post/76928412519>
- [2] Firestone, M. J. & Hedberg, C. W. (2018). Restaurant inspection letter grades and Salmonella infections, New York, New York, USA. *Emerging Infectious Diseases*, 24(12), 2164–2168. <https://doi.org/10.3201/eid2412.180544>

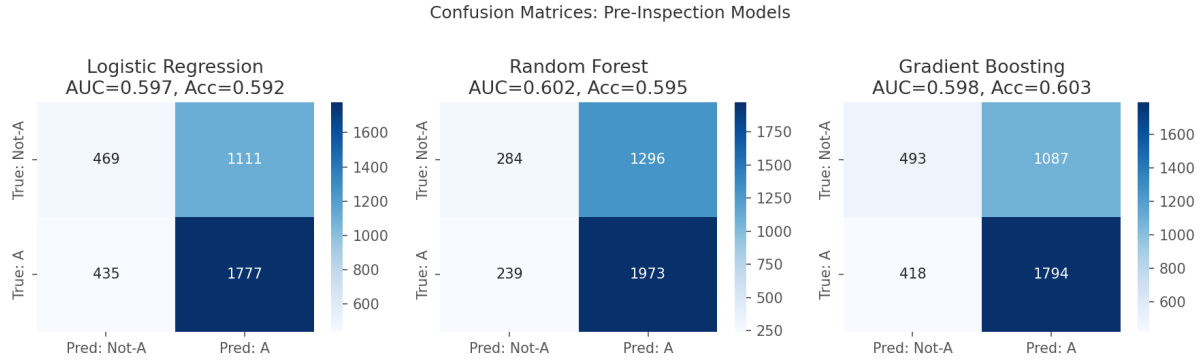


Figure 13: Confusion matrices for the three pre-inspection predictive models. All three models show similar error patterns, consistent with the low AUC.

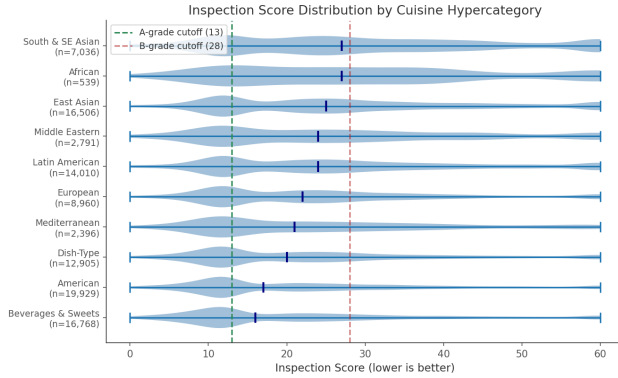


Figure 14: Score distributions by cuisine hypercategory. African and Middle Eastern restaurants tend to score worst; Beverages & Sweets and American establishments score best.